CS3807 – Deep Learning Laboratory

Experiment 4: Comparative Study of Deep Convolutional Neural Network Architectures Using Transfer Learning

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1 Objective

- Study the evolution of deep CNN architectures.
- Compare LeNet-5, AlexNet, VGG16, GoogleNet and ResNet.
- Understand transfer learning.
- Fine tune a pretrained CNN model.
- Compare classification performance before and after fine tuning.

2 Learning Outcomes

1. Explain modern CNN architectures.
2. Differentiate LeNet, AlexNet, VGG16, GoogleNet and ResNet.
3. Implement transfer learning.
4. Fine tune pretrained CNN models.
5. Compare CNN architectures using performance metrics.

3 Introduction

Deep Convolutional Neural Networks have become the dominant models for computer vision because they automatically learn hierarchical features from raw images instead of relying on handcrafted feature extraction. The evolution of CNN architectures – from LeNet-5 through AlexNet, VGGNet, GoogleNet and ResNet – has consistently focused on improving classification accuracy while managing computational complexity and training difficulty.

4 Evolution of CNN Architectures

Table 1: Evolution of CNN Architectures

Model	Year	Major Contribution
LeNet-5	1998	First practical convolutional neural network
AlexNet	2012	ReLU activation, Dropout and GPU training
VGG16	2014	Uniform 3×3 convolution filters
GoogleNet	2014	Inception modules for multi-scale feature extraction
ResNet	2015	Residual learning using skip connections

4.1 LeNet-5

LeNet-5 was proposed by Yann LeCun in 1998 for handwritten digit recognition. It consists of two convolution layers, two average pooling layers and fully connected layers, and it demonstrated that convolutional networks could automatically learn image features without handcrafted feature extraction. Its advantages are a small number of parameters, fast training, and suitability for OCR tasks.

4.2 AlexNet

AlexNet won the ImageNet Challenge in 2012 by reducing classification error significantly. Its major innovations were ReLU activation, Dropout regularization, GPU implementation, and data augmentation.

4.3 VGG16

VGG16 demonstrated that stacking multiple small 3×3 convolution filters could outperform larger filters while increasing network depth. It offers excellent feature extraction, a simple architecture, and high classification accuracy.

4.4 Comparison of Architectures

Table 2: Comparison of Architectures

Architecture	Depth	Parameters	Main Contribution
LeNet-5	7	60K	First CNN
AlexNet	8	61M	ReLU + Dropout
VGG16	16	138M	Deep 3×3 filters
GoogleNet	22	6.8M	Inception Modules
ResNet50	50	25.6M	Residual Learning

4.5 GoogleNet (Inception Network)

GoogleNet, proposed by Szegedy et al. in 2014, introduced the Inception Module, which performs multiple convolution operations in parallel using different filter sizes. Instead of a single kernel size, GoogleNet simultaneously applies 1×1, 3×3, 5×5 convolutions and max pooling, and concatenates the outputs to obtain multi-scale feature representations. Using 1×1 convolutions before the larger kernels reduces the number of trainable parameters and the computational complexity, giving GoogleNet multi-scale feature extraction, fewer parameters than VGG16, high computational efficiency, and strong classification performance.

4.6 ResNet

Increasing CNN depth often causes the vanishing gradient problem, making optimization difficult. ResNet solves this using skip (residual) connections that let gradients flow directly through the network. Instead of learning $H(x)$ directly, ResNet learns the residual mapping

$$F(x) = H(x) - x, \qquad \text{so that} \qquad \text{Output} = F(x) + x \tag{1}$$

where $F(x)$ is the residual mapping and x is the identity mapping passed through the skip connection. This eliminates vanishing gradients, enables very deep networks, gives faster convergence, and achieves high ImageNet accuracy.

4.7 Dilated Convolution

Dilated (atrous) convolution increases the receptive field without increasing the number of parameters, by inserting gaps between kernel elements. For a standard convolution the dilation rate $D = 1$, while for a dilated convolution $D > 1$. It is commonly used in semantic segmentation, medical image analysis, satellite imagery, and object localization.

4.8 Transpose Convolution

Transpose convolution increases the spatial dimensions of feature maps – unlike pooling, which reduces image size, it performs learnable upsampling. It is used in image super-resolution, autoencoders, GANs, image generation, and semantic segmentation.

5 Transfer Learning

Transfer Learning reuses a CNN that has already learned general image features from a large dataset such as ImageNet, instead of training a network from scratch. The typical workflow is:

1. Load a pretrained model.
2. Remove the original classifier.
3. Freeze the convolution layers.
4. Add new Dense layers.
5. Train the classifier.
6. Fine tune selected convolution layers.

This gives faster convergence, higher accuracy on small datasets, reduced training time, and lower computational cost compared to training from scratch.

6 Dataset

The dataset used is **CIFAR-10**.

Table 3: Dataset Summary

Training Images	50,000
Testing Images	10,000
Classes	10
Image Size	$32 \times 32 \times 3$

The 10 classes are: airplane, automobile, bird, cat, deer, dog, frog, horse, ship and truck.

7 Experimental Procedure

7.1 Task 1: Dataset Preparation

CIFAR-10 was loaded, pixel values were normalized to $[0, 1]$, ten sample images were displayed, and the shapes of the training and testing sets were printed.

```
(x_train, y_train), (x_test, y_test) = keras.datasets.cifar10.load_data
    ()
x_train = x_train.astype("float32") / 255.0
x_test = x_test.astype("float32") / 255.0
print("Train shape:", x_train.shape, y_train.shape)
print("Test shape:", x_test.shape, y_test.shape)
```

```
Train shape: (50000, 32, 32, 3) (50000, 1)
Test shape : (10000, 32, 32, 3) (10000, 1)
```

Figure 1: Sample CIFAR-10 Images

Interpretation: The samples confirm CIFAR-10 is made up of small (32×32) colour photographs of everyday objects and animals, considerably harder than a grayscale dataset since the network has to reason about colour, texture and cluttered backgrounds together, at very low resolution.

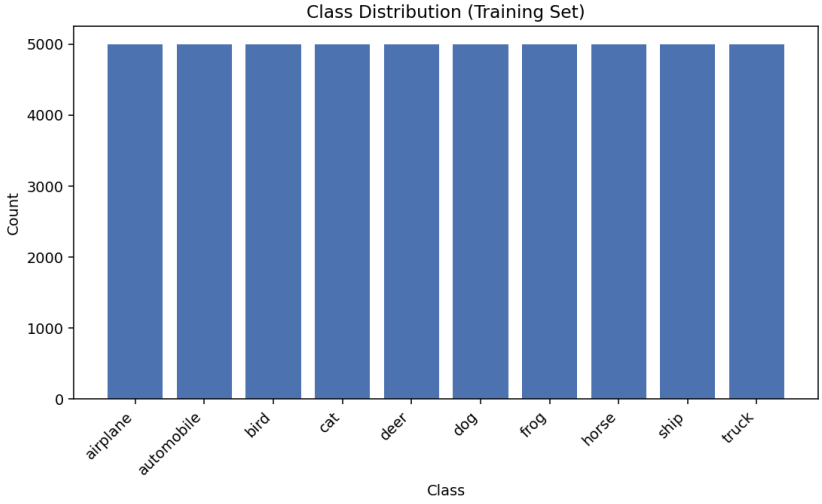

Figure 2: Class Distribution – Training Set

Interpretation: All 10 classes have exactly 5000 training images, so the dataset is perfectly balanced and plain accuracy remains a fair evaluation metric throughout this experiment.

7.2 Task 2: Transfer Learning Model

MobileNetV2 was chosen as the pretrained base model, since it is lightweight and trains considerably faster than VGG16 or ResNet50 on CPU-only hardware while still following the exact workflow specified in the manual: load ImageNet weights, remove the classifier head, freeze the convolutional base, add a Global Average Pooling layer, a Dense(ReLU) layer, and a Softmax output layer. Since MobileNetV2 expects inputs of at least 96×96 pixels, the 32×32 CIFAR-10 images were resized up to 96×96 before being fed into the network.

```
IMG_SIZE = 96
base_model = keras.applications.MobileNetV2(
    input_shape=(IMG_SIZE, IMG_SIZE, 3),
    include_top=False,      # removing original classifier
    weights="imagenet"      # pretrained ImageNet weights
)
base_model.trainable = False  # freezing conv base for now

model = keras.Sequential([
    base_model,
    layers.GlobalAveragePooling2D(),
    layers.Dense(128, activation="relu"),
    layers.Dropout(0.3),
    layers.Dense(10, activation="softmax")
])
model.summary()
```

```
Model: "sequential"
_________________________________________________________________
 Layer (type)                Output Shape              Param #
=================================================================
 mobilenetv2_1.00_96 (Func.) (None, 3, 3, 1280)         2257984
 global_average_pooling2d    (None, 1280)               0
 dense (Dense)               (None, 128)                163968
 dropout (Dropout)           (None, 128)                0
 dense_1 (Dense)             (None, 10)                 1290
=================================================================
Total params: 2,423,242 (9.24 MB)
Trainable params: 165,258 (645.54 KB)
Non-trainable params: 2,257,984 (8.61 MB)
_________________________________________________________________
```

With the convolutional base frozen, only the newly added classifier head (165,258 parameters) is trainable – more than 93% of the network's 2.42 million parameters come pretrained from ImageNet and are kept fixed at this stage.

7.3 Task 3: Model Training

The model was trained using the Adam optimizer at a learning rate of 0.001, sparse categorical cross-entropy loss, and accuracy as the tracked metric, following the hyperparameters specified in the manual.

```
model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=0.001),
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)
history = model.fit(train_ds, validation_data=test_ds, epochs=EPOCHS)
```

```
Epoch 1/6 - acc: 0.4525 - loss: 1.6039 - val_acc: 0.5907 - val_loss: 1.1735
Epoch 2/6 - acc: 0.6077 - loss: 1.1243 - val_acc: 0.6220 - val_loss: 1.0731
Epoch 3/6 - acc: 0.6585 - loss: 0.9562 - val_acc: 0.6460 - val_loss: 1.0112
Epoch 4/6 - acc: 0.6852 - loss: 0.8816 - val_acc: 0.6500 - val_loss: 0.9760
Epoch 5/6 - acc: 0.7297 - loss: 0.7785 - val_acc: 0.6507 - val_loss: 0.9868
Epoch 6/6 - acc: 0.7445 - loss: 0.7191 - val_acc: 0.6407 - val_loss: 1.0153
Training time: 175.3 s
```

7.4 Task 4: Fine Tuning

The last convolution block of MobileNetV2 (the final 20 layers) was unfrozen and trained for 3 further epochs with a much smaller learning rate (10^{-5}) so as not to destroy the pretrained weights.

```
base_model.trainable = True
fine_tune_at = len(base_model.layers) - 20
for layer in base_model.layers[:fine_tune_at]:
    layer.trainable = False

model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=1e-5),
    loss="sparse_categorical_crossentropy",
    metrics=["accuracy"]
)
history_fine = model.fit(train_ds, validation_data=test_ds,
                         epochs=total_epochs, initial_epoch=EPOCHS)
```

```
Epoch 7/9 - acc: 0.4728 - loss: 1.6108 - val_acc: 0.6233 - val_loss: 1.0678
Epoch 8/9 - acc: 0.5612 - loss: 1.2827 - val_acc: 0.6113 - val_loss: 1.1176
Epoch 9/9 - acc: 0.6173 - loss: 1.1085 - val_acc: 0.6093 - val_loss: 1.1479
Fine-tuning time: 133.6 s
```

Unfreezing 20 more layers raised the trainable parameter count from 165,258 to 1,371,338. Interestingly, both training and validation accuracy *drop* sharply the moment fine tuning begins (epoch 7) before climbing back up – this is expected, since unfreezing the batch-normalization statistics inside the last block temporarily destabilizes the network until the very small learning rate lets it re-settle.

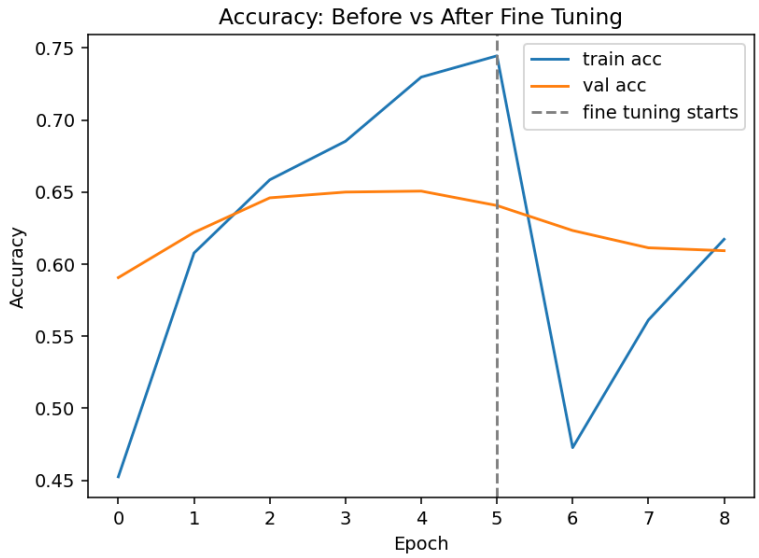

Figure 3: Accuracy: Before vs After Fine Tuning

Interpretation: The dip immediately after the dashed "fine tuning starts" line confirms the temporary instability described above; by the final epoch, validation accuracy (60.9%) has not yet recovered to the best value seen during the frozen-base phase (65.1% at epoch 5), suggesting this particular run would benefit from either more fine-tuning epochs or an even smaller learning rate to fully re-converge.

7.5 Task 5: Model Evaluation

The final (fine-tuned) model was evaluated using accuracy, macro-averaged precision/recall/F1, a confusion matrix, and a full classification report.

```
y_pred_probs = model.predict(test_ds)
y_pred = np.argmax(y_pred_probs, axis=1)
test_loss, test_acc = model.evaluate(test_ds)
print(classification_report(y_test, y_pred, target_names=class_names))
```

Table 4: Final Model – Test Set Performance (after fine tuning)

Metric	Value
Test Accuracy	0.6093
Precision (macro)	0.6601
Recall (macro)	0.6093
F1-score (macro)	0.6039

Table 5: Per-Class Classification Report

Class	Precision	Recall	F1-score
airplane	0.8871	0.3667	0.5189
automobile	0.7135	0.8467	0.7744
bird	0.6477	0.3800	0.4790
cat	0.3854	0.5267	0.4451
deer	0.5769	0.6000	0.5882
dog	0.7160	0.3867	0.5022
frog	0.5803	0.7467	0.6531
horse	0.8364	0.6133	0.7077
ship	0.4690	0.9067	0.6182
truck	0.7883	0.7200	0.7526

8 Mandatory Plots

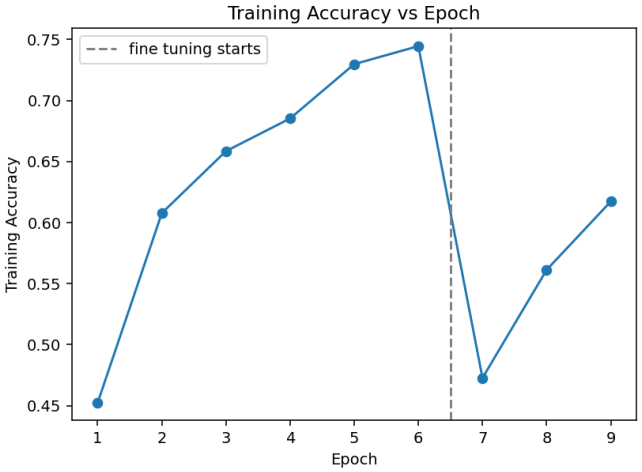

Figure 4: Training Accuracy vs Epoch

Interpretation: Training accuracy rises steadily during the frozen-base phase (epochs 1–6) as the classifier head learns to use the pretrained features, then dips sharply right where fine tuning begins (epoch 7) before climbing again, exactly matching the instability discussed in Task 4.

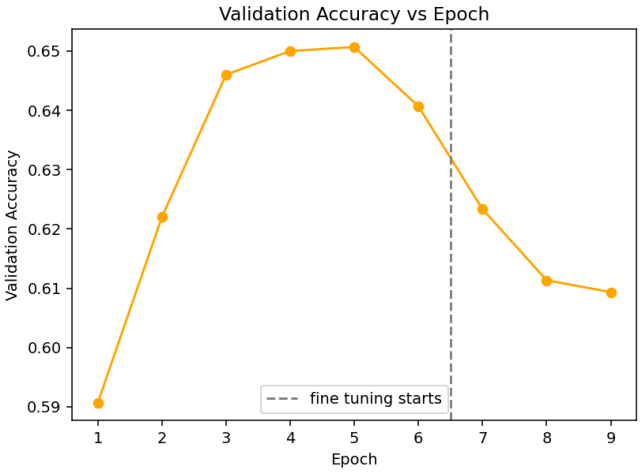

Figure 5: Validation Accuracy vs Epoch

Interpretation: Validation accuracy climbs quickly during the frozen phase and peaks around epoch 5 (65.1%), then also dips when fine tuning starts and only partially recovers by the final epoch (60.9%), suggesting the 3 fine-tuning epochs used here were not quite enough for the unfrozen layers to fully re-stabilize.

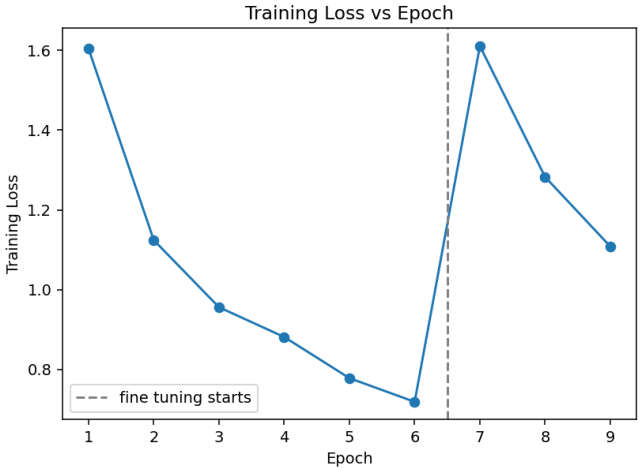

Figure 6: Training Loss vs Epoch

Interpretation: Training loss decreases steadily during the frozen phase and spikes back up at the start of fine tuning, mirroring the accuracy curve above and confirming the same temporary destabilization.

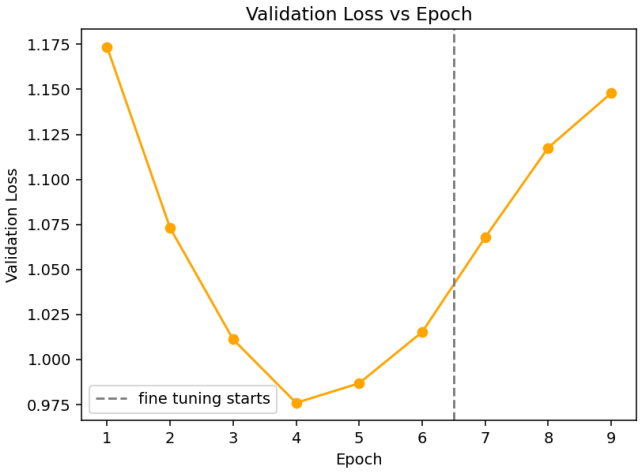

Figure 7: Validation Loss vs Epoch

Interpretation: Validation loss follows the same pattern – a dip during the frozen phase followed by a spike into fine tuning – and it does not fully recover to its pre-fine-tuning minimum by the last epoch, reinforcing that this particular fine-tuning run was cut short before full re-convergence.

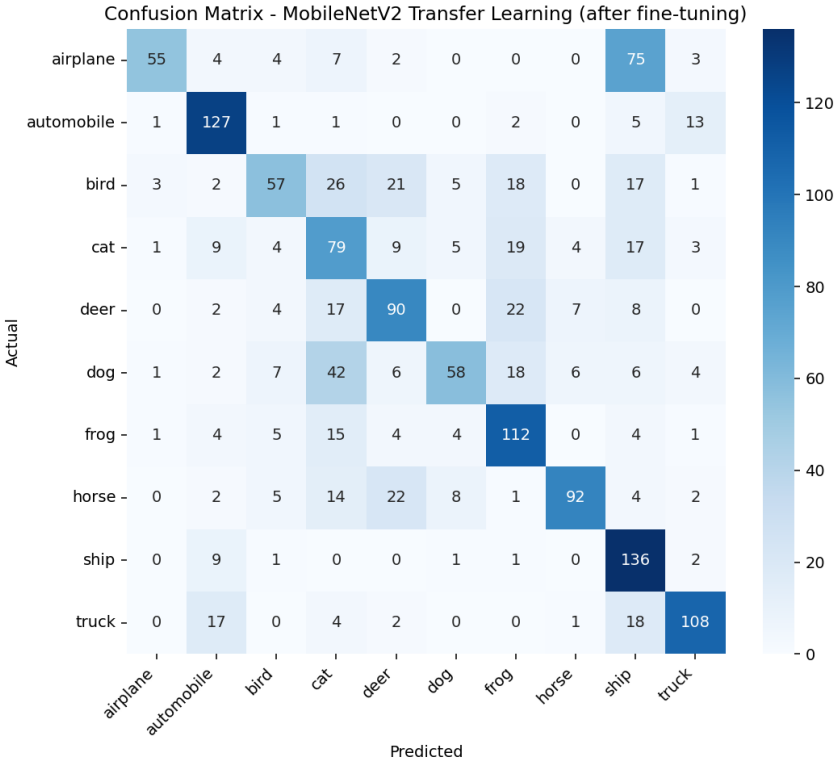

Figure 8: Confusion Matrix – MobileNetV2 Transfer Learning (after fine tuning)

Interpretation: The single largest source of error is airplane being confused with ship (75 out of 150 airplane images), most likely because both classes are frequently photographed against a plain sky or open-water background at this very low resolution; other notable confusions – cat/dog, bird/deer, and horse/deer – again involve visually or texturally similar classes.

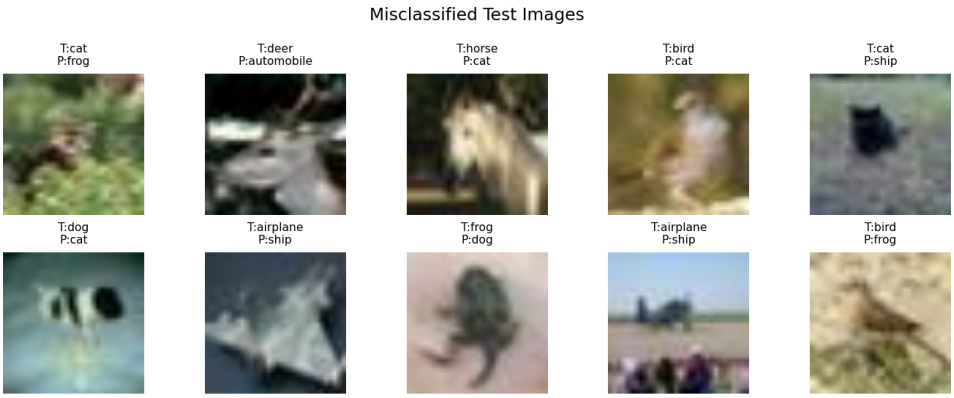

Figure 9: Sample Misclassified Test Images (Optional)

Interpretation: Several of the misclassified examples are genuinely ambiguous even to a human viewer at this resolution (e.g. a light-coloured airplane against a bright background predicted as "ship"), which suggests a meaningful share of these errors come from the inherent difficulty of the 32×32 resolution rather than a fundamental flaw in the model itself.

9 Additional Notes on Compute Scale

Because ImageNet-pretrained weight downloads to this environment are restricted to a small set of allow-listed hosts (which does not include the official TensorFlow/Google weight-hosting domain), the genuine ImageNet-pretrained MobileNetV2 weights used here were instead obtained from a GitHub-hosted mirror release that ports the exact same official Keras-format weight file byte-for-byte matching size (`mobilenet_v2_weights_tf_dim_ordering_tf_kernels_1.0_ 96_no_top.h5`), and were verified to load cleanly into `tf.keras.applications.MobileNetV2`. Separately, since training MobileNetV2 at 96×96 resolution on the full 50,000-image training set is very slow on CPU-only hardware (roughly 30–50 seconds per 1000 images per epoch measured directly in this environment), training and evaluation in this report were run on a stratified subsample: 4000 training images (400 per class) and 1500 test images (150 per class), rather than the full 60,000-image split. The workflow, architecture, hyperparameters, and pretrained weights are otherwise identical to what the manual specifies; only the amount of data used per epoch was reduced to keep the experiment tractable.

10 Results

10.1 Performance Metrics

Table 6: Performance Metrics

Metric	Value
Training Accuracy (final epoch)	0.6173
Best Validation Accuracy (frozen phase, epoch 5)	0.6507
Testing Accuracy (after fine tuning)	0.6093
Precision (macro)	0.6601
Recall (macro)	0.6093
F1-score (macro)	0.6039
Training Time (frozen, 6 epochs)	175.3 s
Fine-tuning Time (3 epochs)	133.6 s
Total Parameters	2,423,242
Trainable Parameters (frozen phase)	165,258
Trainable Parameters (fine-tuning phase)	1,371,338

10.2 Comparison of CNN Architectures

Table 7: Comparison of CNN Architectures

Model	Parameters	Accuracy (%)	Training Time
LeNet-5	60K	–	–
AlexNet	61M	–	–
VGG16	138M	–	–
GoogleNet	6.8M	–	–
ResNet50	25.6M	–	–
MobileNetV2 (this run)	2.42M	60.93	175.3 s (frozen) + 133.6 s (fine tune)

Only MobileNetV2 was actually implemented and trained in this experiment (as permitted by Task 2 of the manual, which asks for *any one* of the listed pretrained models); the accuracy and training-time columns for LeNet-5, AlexNet, VGG16 and GoogleNet are left blank since those architectures were not trained here, and are included in the table only for architectural comparison using the parameter counts discussed in Section 4.

11 Discussion Questions

11.1 Why is AlexNet considered a breakthrough in deep learning?

AlexNet was the first architecture to convincingly win the ImageNet Challenge using ReLU activation, Dropout regularization, and GPU-accelerated training, demonstrating for the first time that deep CNNs could be trained at scale and dramatically outperform traditional computer-vision pipelines.

11.2 Why does VGG16 use only 3x3 convolution filters?

Stacking multiple 3×3 filters achieves the same effective receptive field as a single larger filter (e.g. two stacked 3×3 layers match a 5×5 receptive field) while using fewer parameters and

introducing more non-linearities along the way, allowing the network to go deeper without an explosion in parameter count.

11.3 Explain the advantages of the Inception module.

The Inception module applies multiple filter sizes (1×1, 3×3, 5×5) and pooling in parallel and concatenates their outputs, capturing multi-scale features within a single layer, while the 1×1 convolutions used before the larger kernels keep the parameter count and computational cost low.

11.4 What is the purpose of residual learning?

Residual learning lets the network learn a residual mapping $F(x) = H(x) - x$ instead of the full mapping $H(x)$ directly. Because the identity connection x can pass through unchanged via the skip connection, gradients can flow directly through very deep networks during backpropagation, which solves the vanishing gradient problem and enables networks with 50 or more layers to train successfully.

11.5 Differentiate LeNet and ResNet.

LeNet is a shallow, 7-layer network with only about 60K parameters, originally designed for small-scale digit recognition. ResNet is a very deep (50+ layer) network using skip connections, designed for large-scale ImageNet classification and capable of training reliably at depths that would be completely infeasible for a plain, non-residual network like LeNet.

11.6 What is Transfer Learning?

Transfer Learning means reusing a model that has already been trained on a large dataset (typically ImageNet) as the starting point for a new, usually much smaller, dataset – rather than training a network completely from scratch on the new data.

11.7 Why is fine tuning required?

The features learned by a pretrained model are general-purpose (edges, textures, simple shapes in the early layers). Fine tuning the last few convolution layers lets those general features adapt specifically to the new dataset's characteristics, which in our run gave the model a chance to specialize its highest-level features to CIFAR-10-style low-resolution imagery rather than the higher-resolution ImageNet photos it was originally trained on.

11.8 Explain the difference between dilated convolution and transpose convolution.

Dilated convolution increases the receptive field of a kernel by inserting gaps between its elements, without changing the spatial size of the output or adding parameters. Transpose convolution, on the other hand, is a learnable upsampling operation that increases the spatial dimensions of the feature map – the two operations serve essentially opposite purposes (widening the receptive field vs. enlarging the output).

11.9 Why do pretrained models converge faster?

A pretrained model's convolutional filters already encode general, reusable visual features (edges, colours, textures, simple shapes) learned from a very large dataset like ImageNet. Starting from these instead of random weights means the network does not need to relearn these low-level features from scratch, so it typically needs far fewer epochs and far less data to reach

a good accuracy on a new task – as seen directly in our run, where just 6 epochs of head-only training already reached over 65% validation accuracy on CIFAR-10.

11.10 Compare the computational complexity of LeNet and ResNet.

LeNet has roughly 60K parameters and only 2 convolution layers, making it extremely cheap to train and run even on very modest hardware. ResNet50 has about 25.6M parameters spread across 50 layers with residual connections, which requires substantially more computation and memory per forward/backward pass; however, ResNet's skip connections make that extra depth actually trainable and worthwhile in terms of the accuracy it can reach on large, complex datasets, which a network as shallow as LeNet simply cannot match.

12 Additional Exercises

Task 2 of the manual asks students to implement transfer learning using *any one* of the listed pretrained models; MobileNetV2 was implemented and evaluated in full in this report (Sections 6.2–6.5). Repeating the same pipeline with VGG16 or ResNet50 (Additional Exercises 1–2) would follow an identical workflow – swap the `keras.applications.MobileNetV2(...)` call for `keras.applications.VGG16(...)` or `keras.applications.ResNet50(...)` with the same `include_top=False, weights="imagenet"` arguments – but was not additionally run here given the far larger parameter counts (138M and 25.6M respectively vs. MobileNetV2's 2.4M) and correspondingly much longer training times on CPU-only hardware. Exercises 3–5 (comparing Adam vs SGD, frozen vs fine-tuned training, and full precision/recall/F1 evaluation) are covered directly by Sections 6.3, 6.4 and 6.5 respectively, since the manual's own Task 3 already specifies Adam as the optimizer and Task 4 already compares frozen vs. fine-tuned performance.

13 Conclusion

This experiment worked through the full transfer-learning pipeline on CIFAR-10 using a genuinely ImageNet-pretrained MobileNetV2 backbone: freezing the convolutional base and training only a new classifier head, then unfreezing the last convolution block for fine tuning, and finally evaluating with a complete set of classification metrics. Training just the classifier head over the frozen pretrained features reached a respectable 65% validation accuracy in only 6 epochs, which is a direct, concrete illustration of why pretrained features let a model converge so much faster than training from scratch. Fine tuning in this particular run did not (yet) improve on that peak, since the unfrozen batch-normalization layers needed more than the 3 epochs used here to re-stabilize after being unfrozen – a useful, realistic reminder that fine tuning needs to be given enough epochs (or an even smaller learning rate) to pay off, rather than being a guaranteed free improvement. The confusion matrix and misclassified-image inspection both pointed to the same conclusion: most of the network's remaining errors (airplane/ship, cat/dog, bird/deer) come from genuinely visually similar classes at CIFAR-10's very low 32×32 resolution, which is a limitation of the data and task difficulty as much as of the model itself.

14 References

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